

DSAA3073: Theories in Data Science

Week 2 Open Investigation

When Does a Gradient Update Solve the Operational Problem?

From a battery-dispatch decision to an assumption-sensitive convergence analysis

Basic analysis. A two-battery dispatch problem

Use only the convexity and gradient-method tools developed in the Week 2 Learning Sheet

A municipal microgrid must absorb a short-lived power imbalance. Battery 1 should contribute approximately 3 MW and Battery 2 approximately 1 MW. Because deviation at Battery 2 is operationally more costly, the dispatch centre uses

$$f(w) = \frac{1}{2}((w_1 - 3)^2 + 4(w_2 - 1)^2), \quad w = (w_1, w_2) \in C,$$

with the operating region

$$C = \{w \in \mathbb{R}^2 : |w_1| \leq 4, |w_2| \leq 2\}.$$

The controller begins at $w_0 = (0, 0)$ and proposes the unconstrained gradient update $w_{t+1} = w_t - \eta \nabla f(w_t)$ with $\eta = \frac{1}{4}$. The current software performs one update in constant time for this two-dimensional instance.

Basic Task 1. Establish the geometric and first-order certificate

Prove that C is convex and that f is convex. Derive $\nabla f(w)$, identify the unconstrained minimizer, and use the first-order condition to explain why it is a global minimizer. Check whether it is operationally feasible.

Basic analysis. Audit the proposed controller

A valid update formula is not yet a convergence theorem

Basic Task 2. Execute the update and state the justified decision

Compute w_1, w_2 and the three objective values $f(w_0), f(w_1), f(w_2)$. Check feasibility after each update. On the basis of these calculations, state whether the controller may deploy w_2 , and distinguish the observed two-step decrease from a general convergence guarantee. Finally state how the runtime claim would have to be reformulated if the same calculation were performed for d independently weighted batteries.

Thinking investigation. Formalize the optimization contract

Keep the algorithm visible while changing one modeling assumption at a time

Represent the investigation by

$$(C, f, A_\eta, \mathcal{G}),$$

where C is the feasible region, f the objective, $A_{\eta(w)} = w - \eta \nabla f(w)$ the update rule, and $\mathcal{G}$ the conclusion justified by the stated assumptions. In the basic case, f is a separable positive quadratic with target $b = (3, 1)$.

Investigation 1. Replace the numerical weights by general positive curvature

Let

$$f_{a,b}(w) = \frac{1}{2} \left(a(w_1 - c_1)^2 + b(w_2 - c_2)^2 \right), \quad a, b > 0,$$

and suppose the unconstrained target c is feasible. Determine whether the convexity and global-minimum conclusions survive. For a constant step size, derive a closed-form expression for $w_t - c$ and give the exact condition on η under which every initial point converges to c .

Thinking investigation. Remove convexity

A stationary point may persist while the theorem disappears

Investigation 2. Reverse the curvature in one operational direction

For $\kappa > 0$, replace the objective by

$$q_{\kappa(w)} = \frac{1}{2} \left((w_1 - 3)^2 - \kappa(w_2 - 1)^2 \right).$$

Decide whether the convex first-order theorem still applies and construct a direct witness if it does not. The gradient is still zero at $(3, 1)$; decide whether this point is a minimizer. Derive the exact gradient-descent error sequence and identify the exceptional initial conditions for which the unstable coordinate does not diverge.

Thinking investigation. Allow the step size to vary

The constant geometric factor becomes an analytic product

Investigation 3. Replace one constant step size by a schedule

Return to $f_{a,b}$ but use positive step sizes $\eta_0, \eta_1, \dots$. Determine whether the constant-step convergence formula remains valid. Derive the exact expression for each error coordinate after t updates and state a necessary-and-sufficient product condition for convergence from every initial point. Give a simple sufficient condition that is easy to verify operationally.

Synthesis. State the transferable conclusion

Return to the microgrid. Identify which claims concern geometry, which concern optimization dynamics, and which concern computation. State what must be checked before the same controller is deployed with new physical weights or a new step-size schedule.